| Course Code PHY6002 | Solar – Terrestrial Physics | Course Type | LT |
|---|---|---|---|
| | | Credits | 3 |

**Course Objectives:** Aims to equip and assist students to gain advanced knowledge and skills in the study of Solar Terrestrial Physics and Ionospheric study techniques

**Course Outcomes:**

CO1 - To understand the characteristics of our nearest star; The Sun and its transient phenomenon.

CO2 - To understand diagnostic techniques for analyzing solar phenomena.

CO3 - To understand interplanetary medium and Geospace.

CO4 - To understand Earth's atmosphere and its diagnostic techniques.

| Mod. No. | Module Description | Lect. Hrs | CO |
|---|---|---|---|
| 1 | **The Sun and its Atmosphere**<br><br>The Sun; composition, structure, motion, magnetic field. Solar Atmosphere; layers - photosphere, chromosphere, transition-region and corona, characteristics - temperature, density, and magnetic structures - sunspots, plage, facule, coronal hole. Quiet Sun. | 3 | 1 |
| 2 | **Solar Transients and their Genesis**<br><br>Solar Transients; jet-like structures, solar flares, coronal mass ejections (CMEs), filaments, solar wind, solar energetic particles (SEPs). Basics of physical processes in solar plasma; magnetic reconnection, plasma flows, thermal emission process, non-thermal emission process, coronal heating problem, spectral, magnetic and kinematic characteristics. | 9 | 1 |
| 3 | **Diagnostics of Solar Plasma**<br><br>Solar Spectrum; emission line, absorption lines, and continuum, fundamentals of line formation. Spectroscopic diagnostics of the solar plasma by its radiation; multi-thermal emissions, x-ray, extreme ultraviolet (EUV), ultraviolet (UV), optical and radio. Physical conditions in the solar atmosphere, elementary processes and mechanisms of radiation formation, and methods and results of the temperature and density diagnostics of various structures (such as active and quiet solar regions, coronal holes, and the extended corona). | 12 | 2 |

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| **4** | **Interplanetary Medium and Geospace**<br>Definition and characteristics (general) of interplanetary medium; magnetic field, plasma flow - solar wind, interplanetary CMEs, CIRs, cosmic rays. Earth's Magnetosphere, Solar Wind - Magnetosphere interaction, Bow Shock, Radiation Belts, Aurora, Geomagnetic Storms; distrubed magnetic fields, ring current, Geomagnetic Indices, Dst, AE, AU, PC, Ap, Kp | **9** | **3** |
| **5** | **Earth's Atmosphere and Space Weather**<br>Earth's Atmosphere; layers, height -density profiles. Ionosphere- D, E, F layers; electron density, ion density, characteristic frequencies. Ionospheric response to Solar Transients. Physical processes, plasma flow, current sheet formation, ionospheric wind, fountain effect, scintillation. Ground and Satellite based observations of ionosphere; Ionosonde, GPS, characteristic parameters - TEC, foF2, fmin, hmF2, NmF2 etc. Case Study on Space Weather. | **12** | **4** |
| | **Total Lecture Hours:** | **45** | |

**Mode of Teaching and Learning:**
Flipped Class Room, Activity Based Teaching/Learning, Digital/Computer based models, wherever possible to augment lecture for practice/tutorial and minimum 2 hours lectures by industry experts on contemporary topics

**Mode of Evaluation:**
The assessment and evaluation components may consist of unannounced open book examinations, quizzes, assignments, tutorials and any other innovative assessment practices followed by faculty, in addition to the Continuous Assessment Tests and Term End Examination.

**Text Book(s):**

- **Advanced Magnetohydrodynamics** by J. P. Goedbloed; Rony Keppens; Stefaan Poedts
- **Introduction to Space Physics** by Margaret J Kivelson and Christopher T .Russel
- **An Introduction to Space Weather** by Mark Moldwin

**Reference Books:**

- **Continuous Measurements of the Virtual Heights of the Ionosphere** by Gilliland TR
- **GPS/GNSS: Principles and Applications** by Alberto Cina, Excelic Press

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| **Recommendation by the Board of Studies on** | 11th March 2022 |
| **Approval by Academic council on** | |
| **Compiled by:**<br>**Dr. Pradeep Kumar Kayshap**<br>**Dr. Sharad Chandra Tripathi**<br>**Dr. Shweta Mukherjee** | |